INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR

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TOPIC: Software Fault Prediction using Fuzzy Logic

Spring (2025 – 2026)

Submitted by-

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Introduction

This project presents a phased Software Fault Prediction model utilizing a **Fuzzy Inference System (FIS)**. Predicting software faults early in the development lifecycle significantly enhances quality control and reliability. Instead of relying on a single-stage prediction, this model employs a multi-stage, cascaded Mamdani Type-1 FIS that evaluates software metrics across three critical phases: Requirements, Design, and Coding. Here we'll do the following-

Select dataset → Choose metrics → Define inputs/outputs → Design membership functions → Create fuzzy rules → Build FIS → Perform inference → Defuzzification → Analyze results → Evaluate performance → Draw conclusions

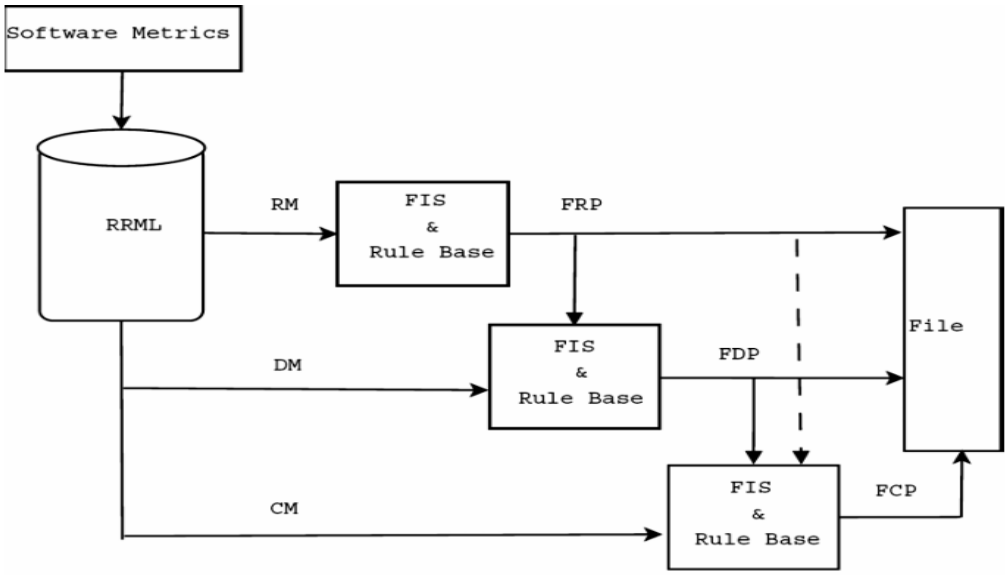


Fig 1: System Architecture

Dataset Source

The dataset used in this study is taken from the reference material provided by our professor.The values of software quality metrics such as RCR, RIW, PM, DDD, FDN, DC, CDD, and CC, along with their respective ranges and classifications, are adopted from the instructor's dataset.

These values represent realistic software engineering scenarios and are used to design membership functions and fuzzy rules for fault prediction.

Since the dataset already contains structured metric ranges and fault classifications, it is directly utilized to build and evaluate the fuzzy inference system.

The architecture consists of three interconnected Fuzzy Inference Systems:

1. **Phase 1 (Requirements Phase):** Evaluates Requirements Metrics (RM) to calculate the Faults in Requirements Phase (FRP).
2. **Phase 2 (Design Phase):** Evaluates Design Metrics (DM) along with the inherited FRP to calculate the Faults in Design Phase (FDP).
3. **Phase 3 (Coding Phase):** Evaluates Coding Metrics (CM) along with both FRP and FDP to determine the overall Faults in Coding Phase (FCP).

RRML = Repository of Requirement Metrics Library, A **data storage / database** that contains all our software metrics like (RCR, RIW, DDD etc.)

Use of Decision Tree Algorithm

Decision trees require precise boundaries and crisp data, whereas software metrics often involve uncertainty and imprecision.

Fuzzy logic is better suited for this problem because it can handle linguistic variables such as "Low", "Moderate", and "High", and model gradual transitions between them. Therefore, a Fuzzy Inference System (FIS) was chosen as the primary modeling technique for improved interpretability and flexibility.

Phase-wise Software Metrics and Risk Level (Low to High) Classification

Phase NO	Phase name		Rank	Ranges	Low	Moderate	High
1	Requirement	RCR	0.7	[0-100]	[0 0 25 35]	[25 35 55 70]	[60 80 100 100]
		RIW	0.61	[0-5]	[0 0 2]	[1 2 3]	[3 5 5]
		PM	0.60	[0-5]	[0 0 0.5 1]	[1 1.5 2.5 3]	[3 3.5 5 5
2	Design	DDD	0.75	[0-1]	[0 0 0.2 0.4]	[0.2 0.4 0.5 0.7]	[0.5 0.7 1 1]
		FDN	0.71	[0-50]	[0 0 2 5]	[1 4 8 15]	[8 20 50 50]
		DC	0.62	[0-500]	[0 0 50 100]	[80 120 200 250]	[200 300 500 500]
3	coding	CDD	0.83	[0-1]	[0 0 0.2 0.4]	[0.2 0.4 0.5 0.7]	[0.5 0.7 1 1]
		CC	0.74	[0-500]	[0 0 100 150]	[100 150 250 300]	[250 350 500 500]

Output Variable Classification for Fault Prediction (Very Low to Very High)

Phase NO	Phase name	O/P variables	Ranges	Very Low	Low	Moderate	High	Very high
1	Requirement	FRP	[0-100]	[0 0 5]	[2.5 5 7]	[5 10 20]	[15 30 45]	[30 100 100]
2	Design	FDP	[0-100]	[0 0 10]	[2 10 15]	[10 20 30]	[20 40 60]	[50 100 100]
3	coding	FCP	[0-100]	[0 0 15]	[5 15 30]	[15 35 60]	[30 55 80]	[60 100 100]

Fault Prediction Model Structure (Input–Output Mapping by Phase)

Phase	I/P	O/P	Full name
1	RCR+RIW+PM	FRP	Fault Risk Prediction
2	FRP+DDD+FDN+DC	FDP	Fault Density Prediction
3	FRP+FDP+CDD+CC	FCP	Fault Count Prediction

Procedure:

The following steps are followed for designing the system in online matlab:

Step 1: Open Fuzzy Logic Designer from apps.

Step 2: *Define Inputs and Outputs:* Specify the number of inputs and outputs
Step 3: *Configure Input Variables:* Assign names, ranges, shape, parameters

Step 4: *Configure Output Variables:* Assign names, ranges (add MF to increase output ranges from Very low to very high), shape, parameters

Step 5: *Analyze FIS and Membership Functions:* Observe the FIS structure and verify the membership functions using the FIS Plot and Membership Function Editor to ensure proper overlap and coverage of the input space.

Step 6: *Define Fuzzy Rules:* Fuzzy rules are created using the Rule Editor:
- 27 rules for FRP (Requirement phase)
- 20–50 rules for FDP (Design phase)
- 15–25 rules for FCP (Coding phase)

Instead of using all possible combinations, a reduced rule base is used to improve interpretability and avoid redundancy

Step 7: *Perform Rule Inference:* The Rule Inference system is used to evaluate how input values activate different rules and contribute to the final output.

Step 8: *Analyze Control Surface:* Control surface plots are generated to visualize the relationship between inputs and outputs. The plots help in understanding how changes in input variables affect the predicted number of faults. From the control surface, it is observed that increasing input complexity or defect-related metrics results in higher fault prediction values.

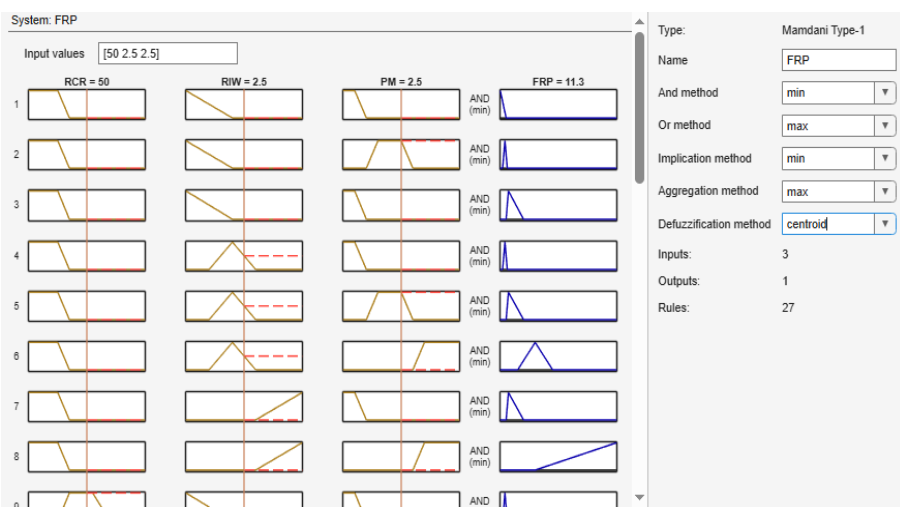
Step 9: Result

Defuzzification Method:

Centroid method is used for Defuzzification . It calculates the center of gravity of the output fuzzy set. As we are using Continous values, so we can use the following formula

$$Z = \int \mu(z) \cdot z \, dz \, / \int \mu(z) \, dz$$

Fig 2: Defuzzification of FRP(11.3)



MATLAB Implementation:

From Phase 1:

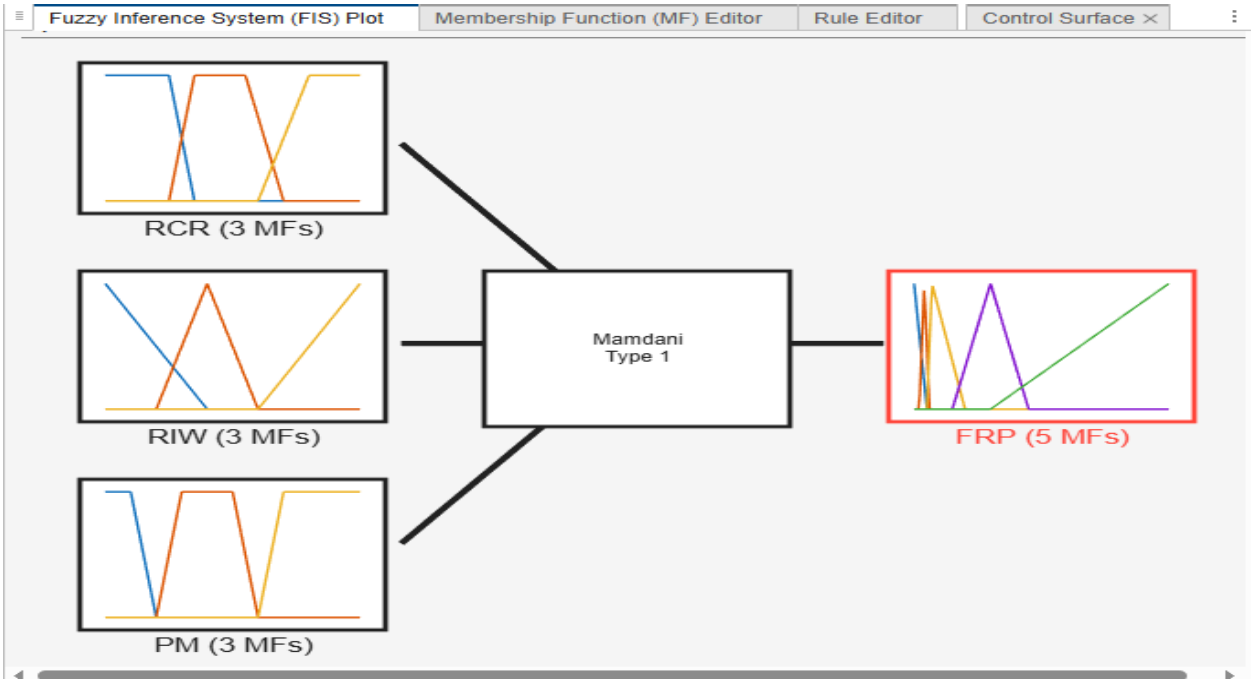


Fig: Fuzzy Inference System (FIS) Plot of Phase 1

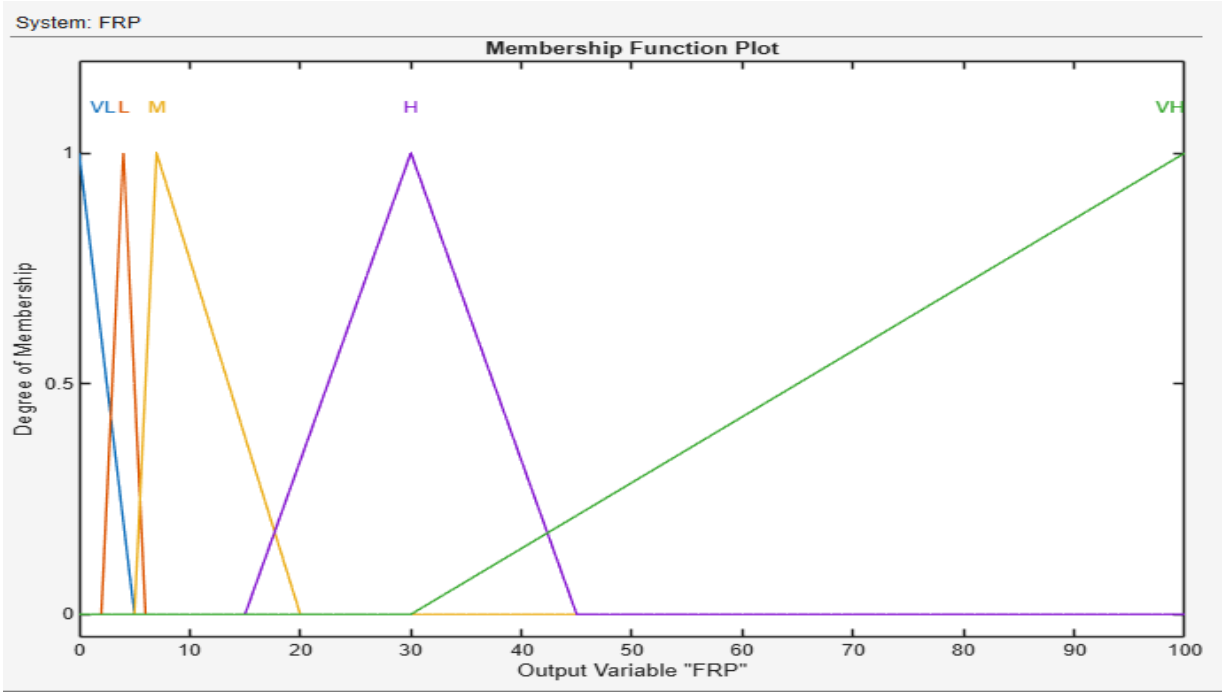


Fig: Membership Function (MF) plot of FRP

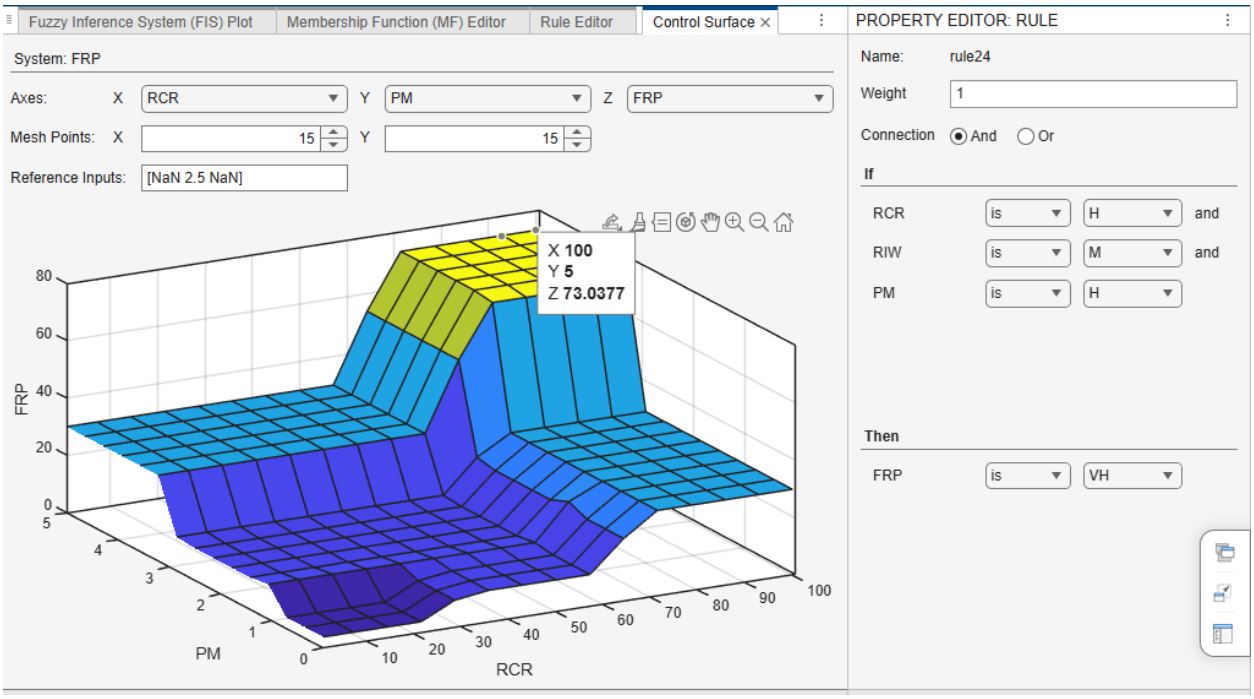


Fig: Control Surface diagram of phase 1

X axis RCR is high (100) and Y axis PM is high (5), Z axis the value of FRP is maximum (around 73), indicating a very high number of faults.

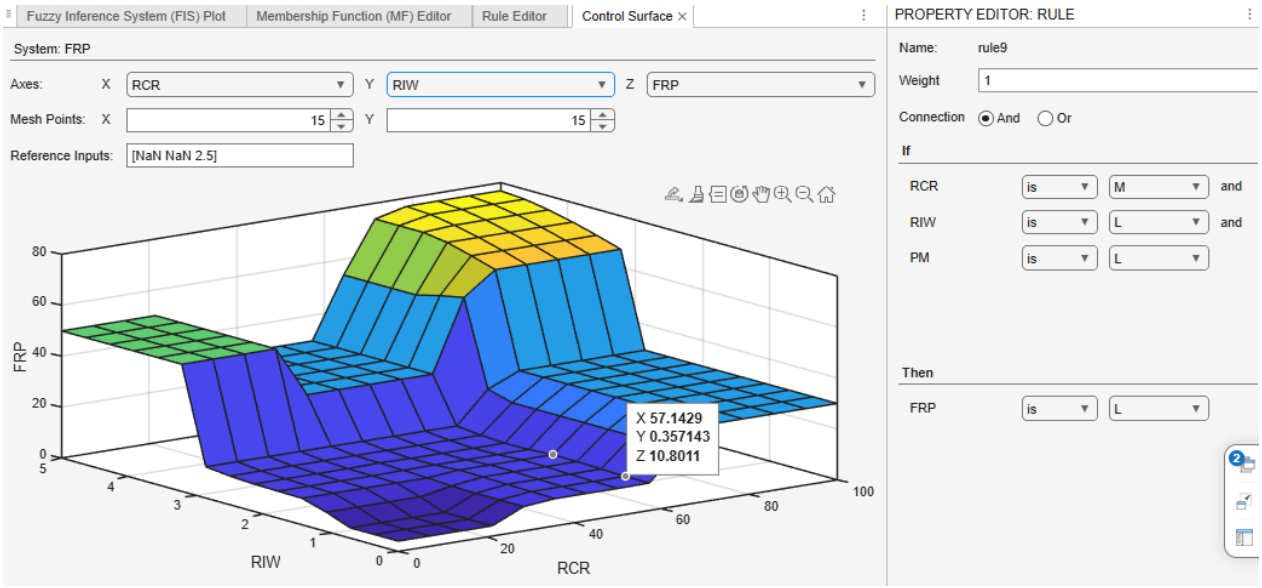
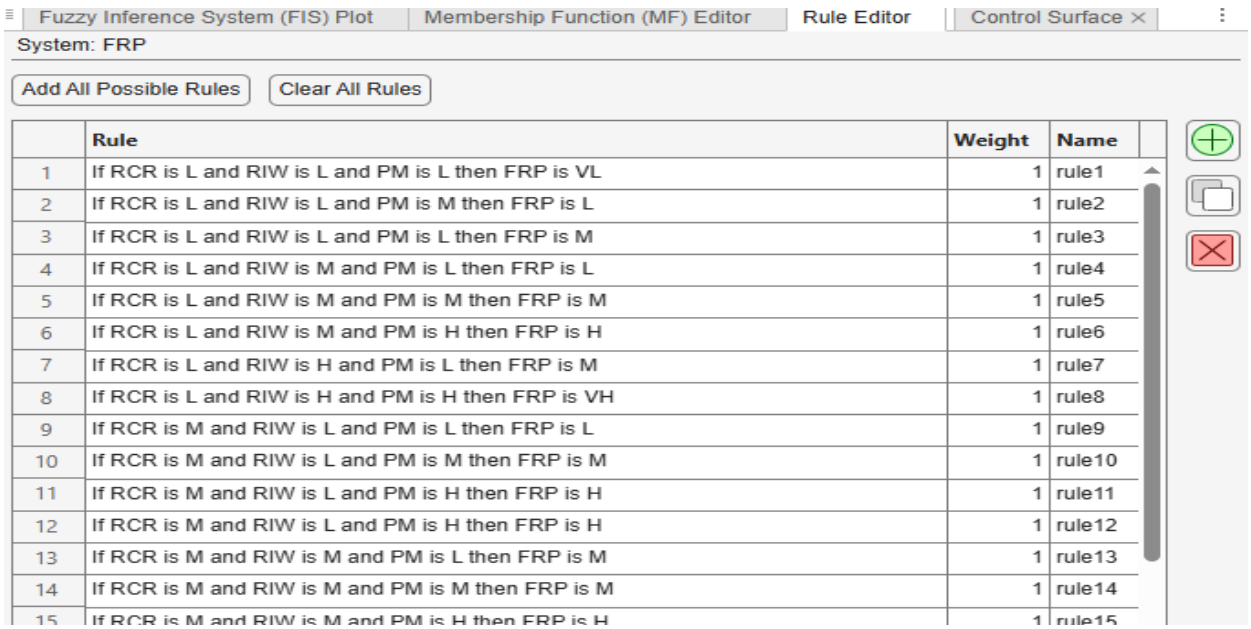


Fig: Control Surface diagram of phase 1

X axis RCR is moderate (around 57) and Y axis RIW is low (0.36), Z axis the value of FRP is low (around 10), indicating a less number of faults.

Observation from control surface:

- When **RCR (Requirement Change Rate) is high**, FRP rises sharply, indicating strong sensitivity to requirement changes.
- **PM (Project Metrics)** also contributes positively, but its effect is **moderate compared to RCR**.
- At **high RCR and high PM**, FRP reaches **Very High levels (~70–80)**.
- When both inputs are low, FRP remains **very low**, confirming correct model behavior.
- The surface shows a **steep rise region**, indicating **threshold-based risk escalation**.
- **RCR is the dominant factor**, strongly influencing FRP across all regions.
- **RIW (Requirement Importance Weight)** acts as a **supporting factor**, increasing FRP gradually.
- When **RIW is low and RCR is moderate**, FRP stays **low to moderate (~10–20)**.
- At **high RCR + high RIW**, FRP becomes **high to very high**, showing combined effect.
- The smooth surface indicates **stable fuzzy rule transitions**



Fuzzy Inference System (FIS) Plot | Membership Function (MF) Editor | Rule Editor | Control Surface

System: FRP

Add All Possible Rules | Clear All Rules

	Rule	Weight	Name
1	If RCR is L and RIW is L and PM is L then FRP is VL	1	rule1
2	If RCR is L and RIW is L and PM is M then FRP is L	1	rule2
3	If RCR is L and RIW is L and PM is L then FRP is M	1	rule3
4	If RCR is L and RIW is M and PM is L then FRP is L	1	rule4
5	If RCR is L and RIW is M and PM is M then FRP is M	1	rule5
6	If RCR is L and RIW is M and PM is H then FRP is H	1	rule6
7	If RCR is L and RIW is H and PM is L then FRP is M	1	rule7
8	If RCR is L and RIW is H and PM is H then FRP is VH	1	rule8
9	If RCR is M and RIW is L and PM is L then FRP is L	1	rule9
10	If RCR is M and RIW is L and PM is M then FRP is M	1	rule10
11	If RCR is M and RIW is L and PM is H then FRP is H	1	rule11
12	If RCR is M and RIW is L and PM is H then FRP is H	1	rule12
13	If RCR is M and RIW is M and PM is L then FRP is M	1	rule13
14	If RCR is M and RIW is M and PM is M then FRP is M	1	rule14
15	If RCR is M and RIW is M and PM is H then FRP is H	1	rule15

Fig: Rule Base for Requirement Phase – Fuzzy Rules Mapping (RCR, RIW, PM) to FRP

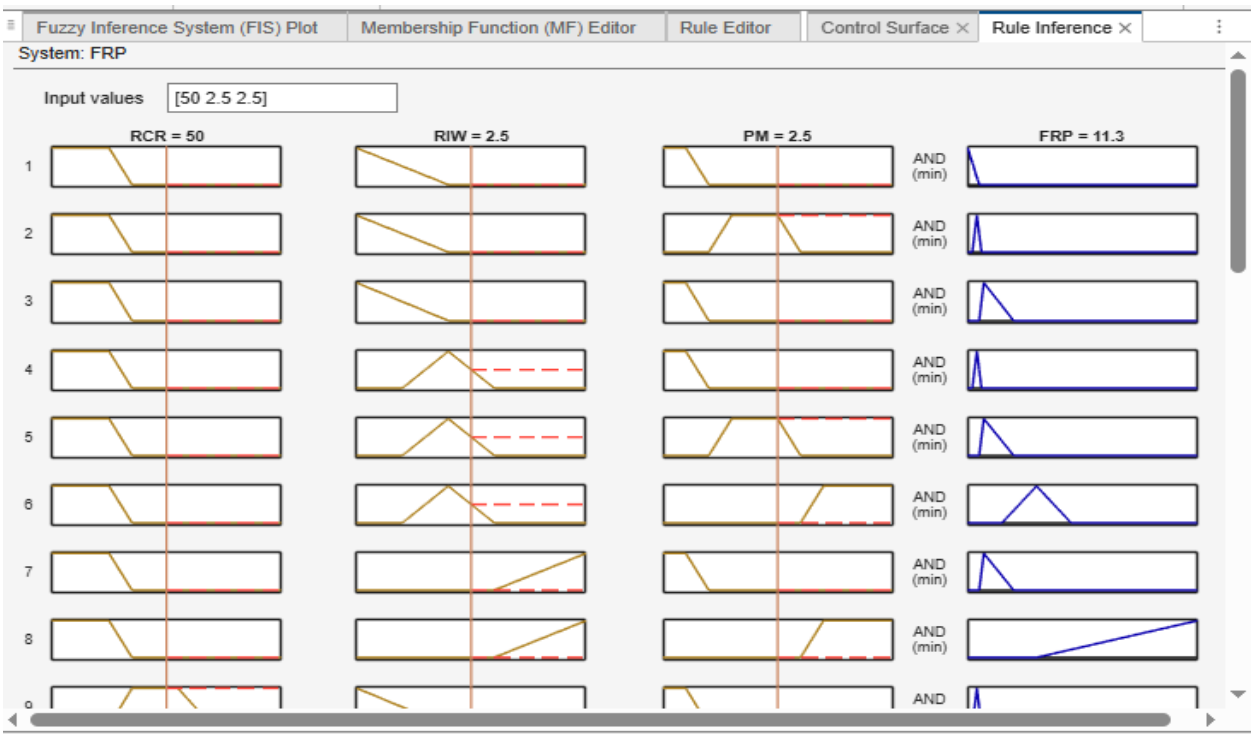


Figure: Fuzzy Inference System Rule Viewer Showing FRP Output for Given Inputs

From Phase 2:

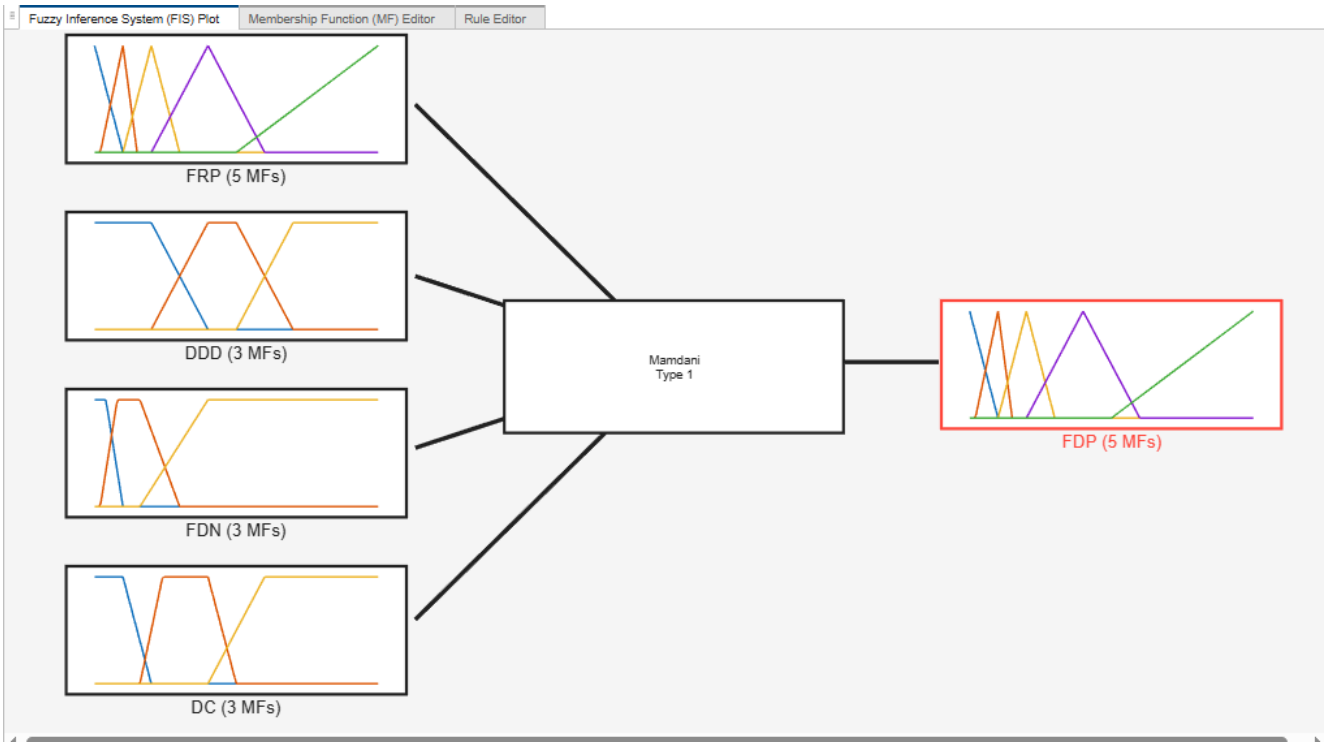


Fig: Fuzzy Inference System (FIS) Plot of Phase 2

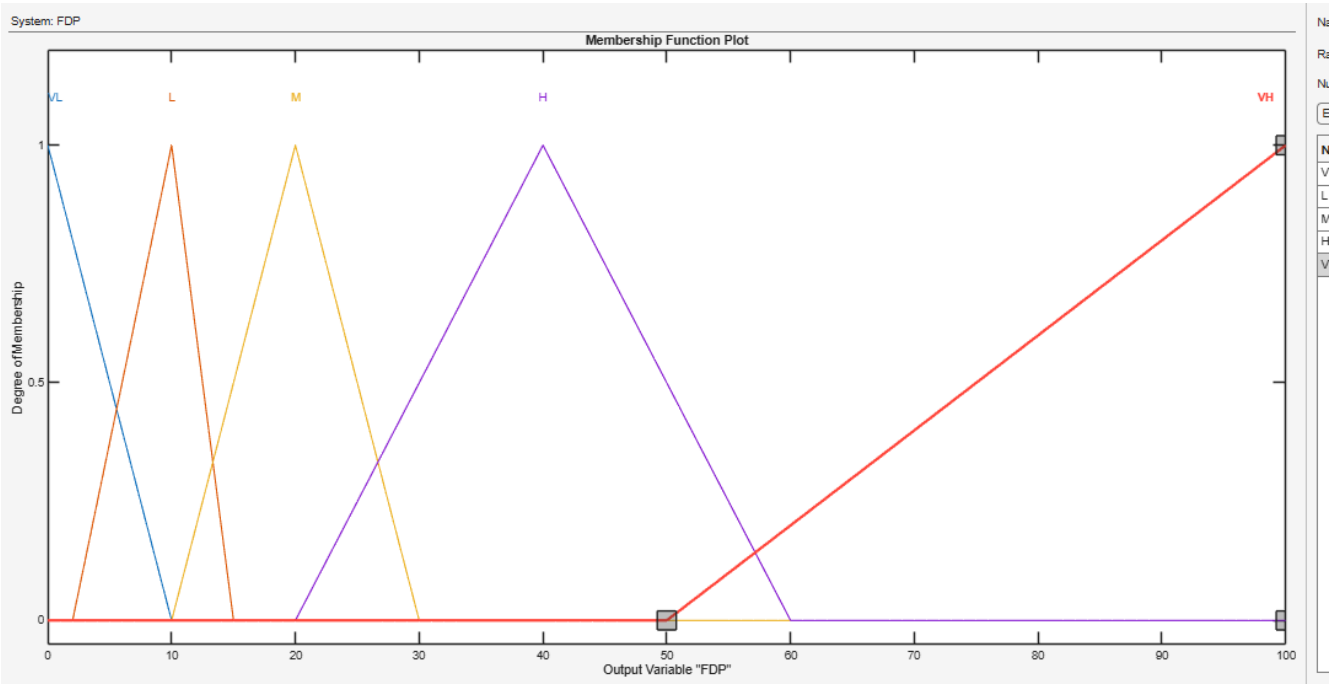


Fig: Membership Function (MF) plot of FDP

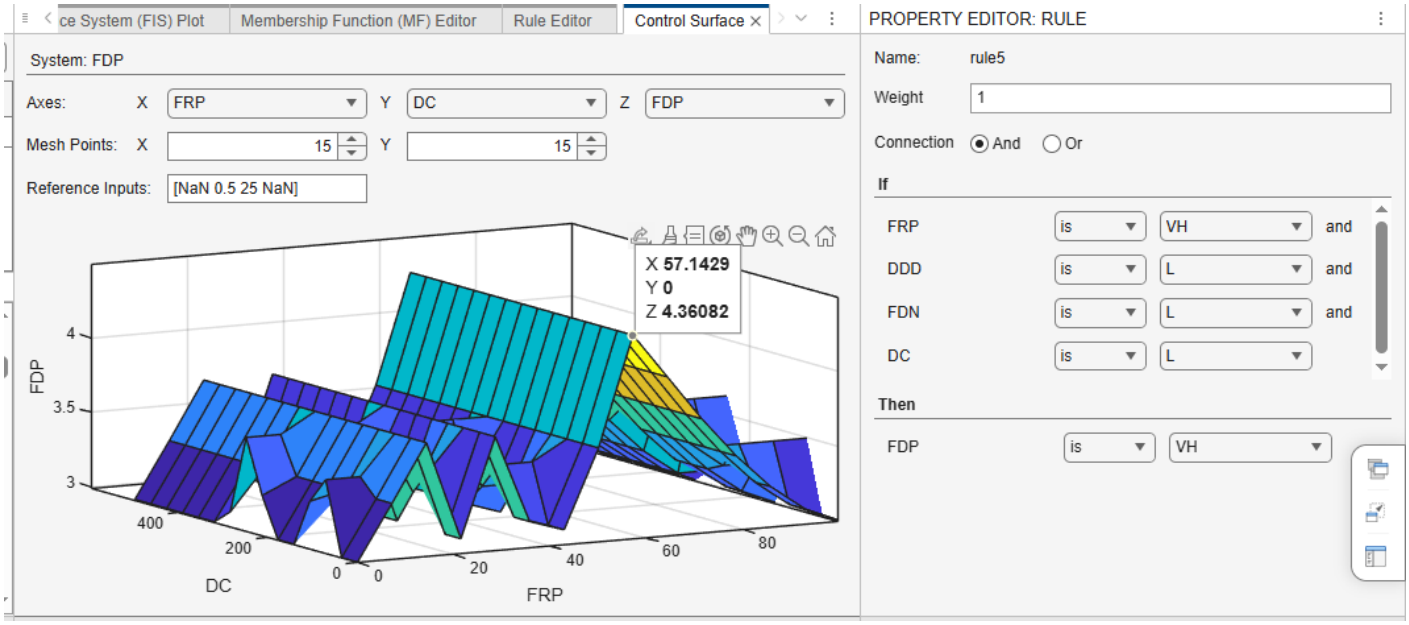


Fig: Control Surface Diagram of phase 2

X axis FRP is high (around 57) and Y axis DC is low (0), Z axis the value of FDP is high (around 4), indicating a high number of fault density prediction

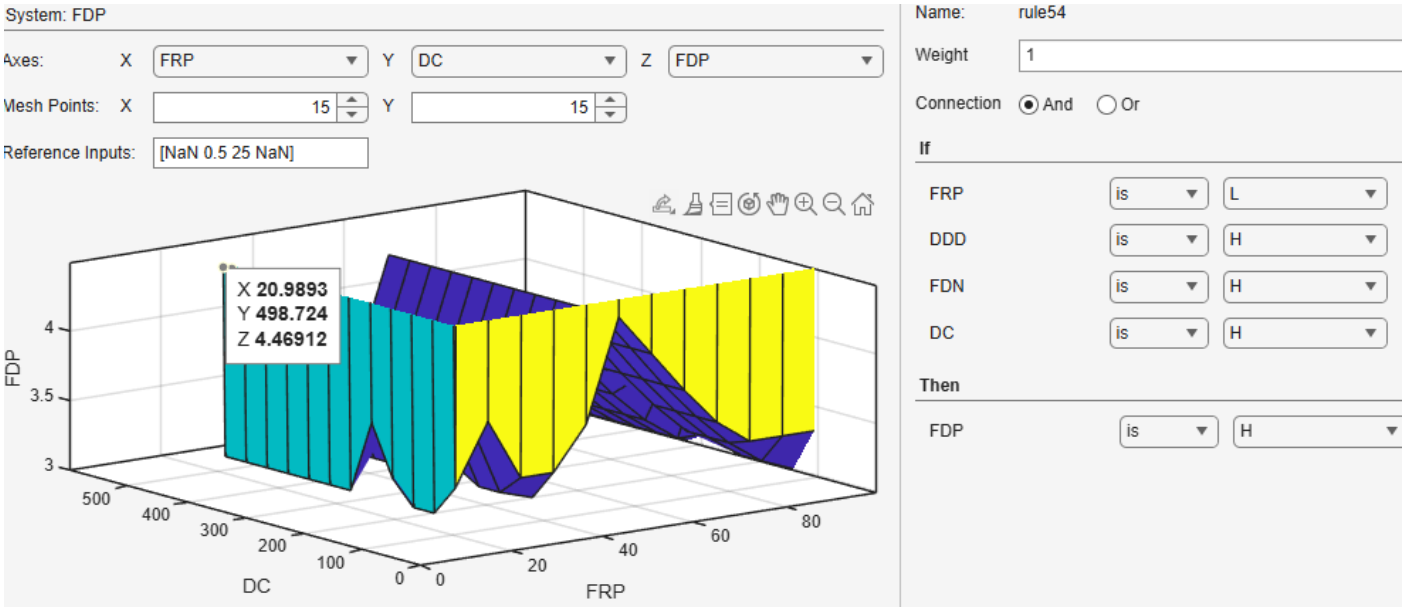


Fig: Control Surface Diagram of phase 2

X axis FRP is low (around 20) and Y axis DC is high (498), Z axis the value of FDP is high (around 4.47), indicating a high number of fault density prediction

Observation from control surface:

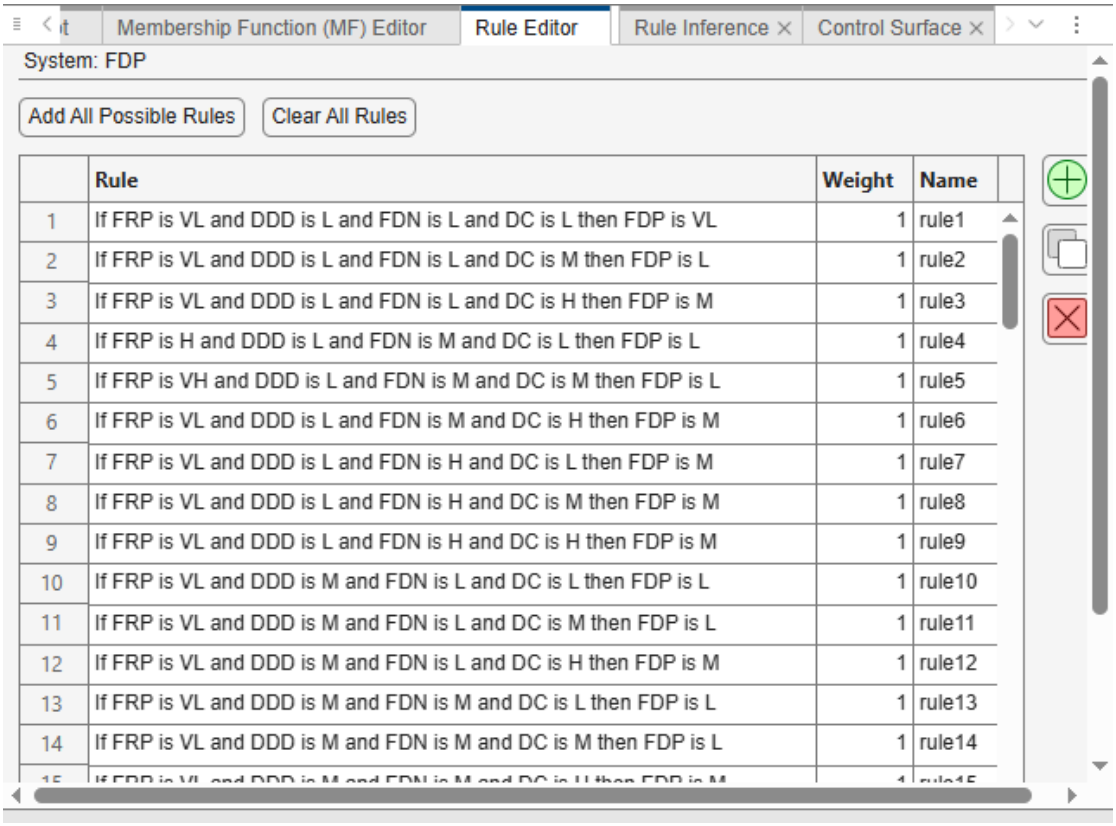
1. When **FRP is high, FDP values increase** significantly, indicating that faults from the requirement phase propagate into the design phase.

2. As **DC increases, FDP also increases**, showing that higher design complexity introduces more defects.

3. When **both FRP and DC are high, FDP reaches its maximum** values around 4–5 in the scaled range), indicating a high fault-prone design.

4. When **FRP is low, FDP remains** low even if DC increases, suggesting that good requirement quality reduces design faults.

5. The surface is non-linear, showing that the combined effect of FRP and DC is not purely additive, which demonstrates the effectiveness of fuzzy logic.



	Rule	Weight	Name
1	If FRP is VL and DDD is L and FDN is L and DC is L then FDP is VL	1	rule1
2	If FRP is VL and DDD is L and FDN is L and DC is M then FDP is L	1	rule2
3	If FRP is VL and DDD is L and FDN is L and DC is H then FDP is M	1	rule3
4	If FRP is H and DDD is L and FDN is M and DC is L then FDP is L	1	rule4
5	If FRP is VH and DDD is L and FDN is M and DC is M then FDP is L	1	rule5
6	If FRP is VL and DDD is L and FDN is M and DC is H then FDP is M	1	rule6
7	If FRP is VL and DDD is L and FDN is H and DC is L then FDP is M	1	rule7
8	If FRP is VL and DDD is L and FDN is H and DC is M then FDP is M	1	rule8
9	If FRP is VL and DDD is L and FDN is H and DC is H then FDP is M	1	rule9
10	If FRP is VL and DDD is M and FDN is L and DC is L then FDP is L	1	rule10
11	If FRP is VL and DDD is M and FDN is L and DC is M then FDP is L	1	rule11
12	If FRP is VL and DDD is M and FDN is L and DC is H then FDP is M	1	rule12
13	If FRP is VL and DDD is M and FDN is M and DC is L then FDP is L	1	rule13
14	If FRP is VL and DDD is M and FDN is M and DC is M then FDP is L	1	rule14

Fig: Rule Base for Requirement Phase – Fuzzy Rules Mapping (FRP, DDD, FDN, DC) to FDP

Phase 3:

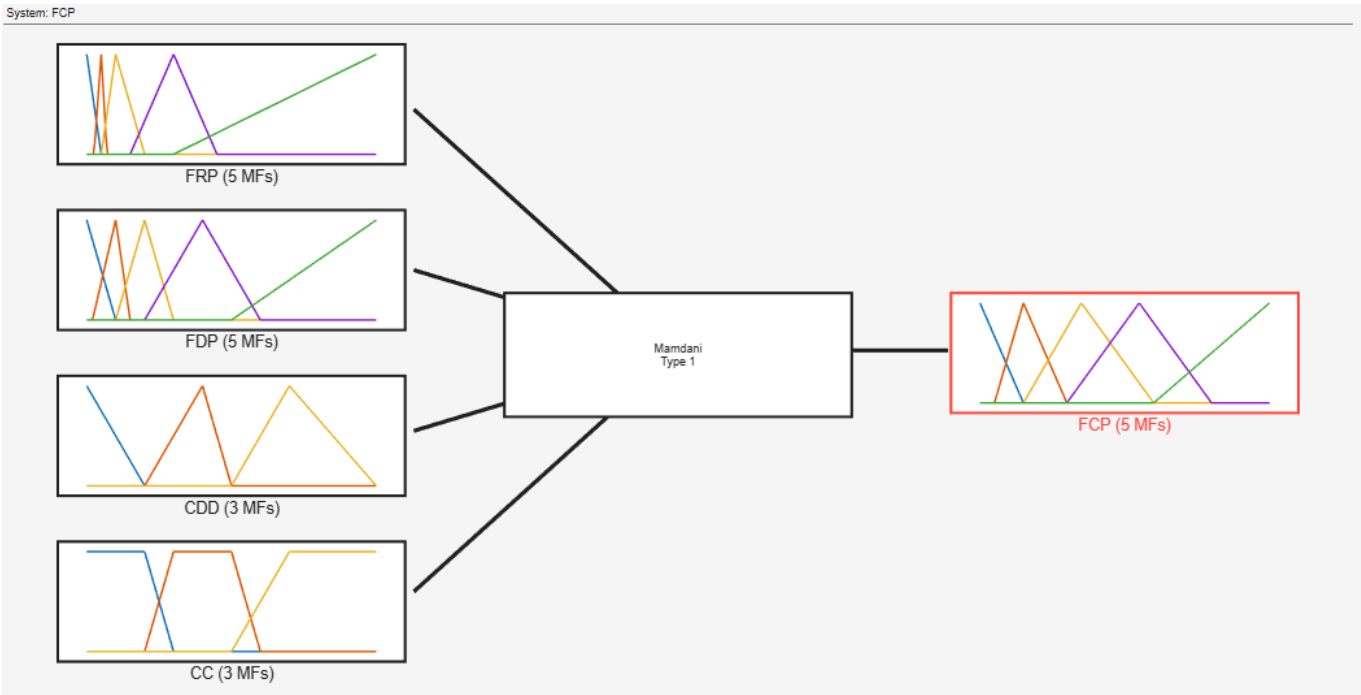


Fig: Fuzzy Inference System (FIS) Plot of Phase 3

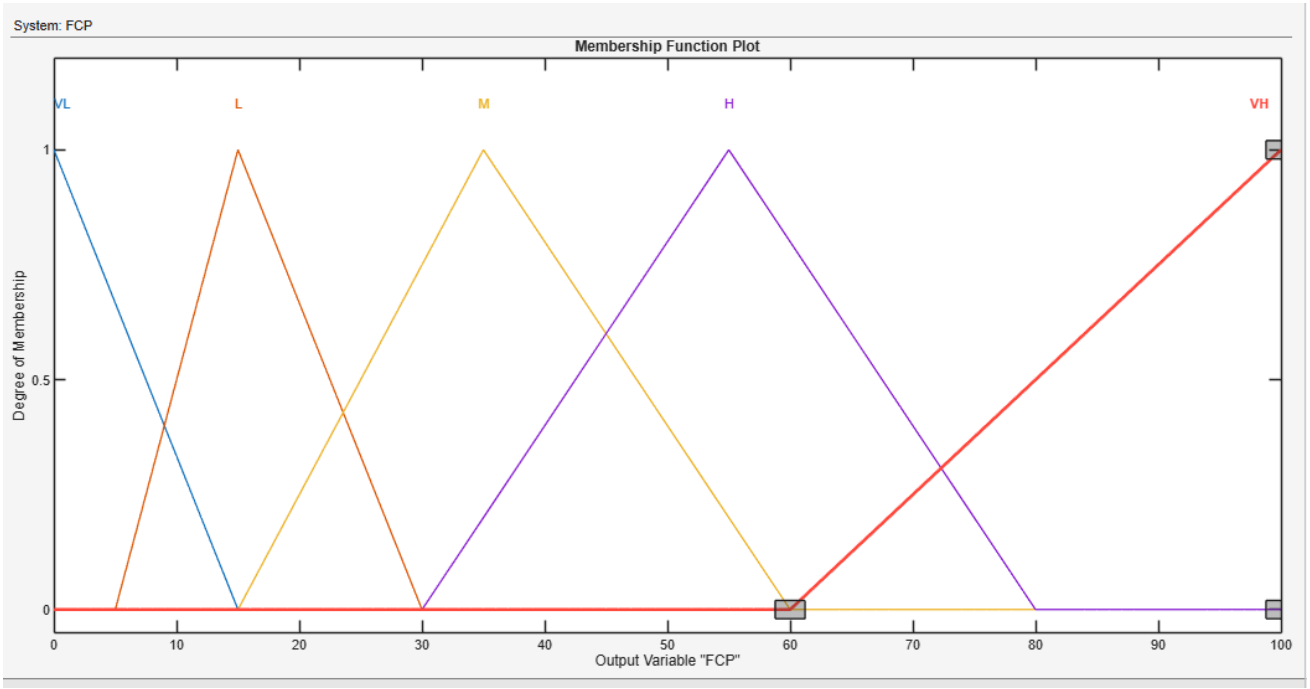


Fig: Membership Function (MF) plot of FCP

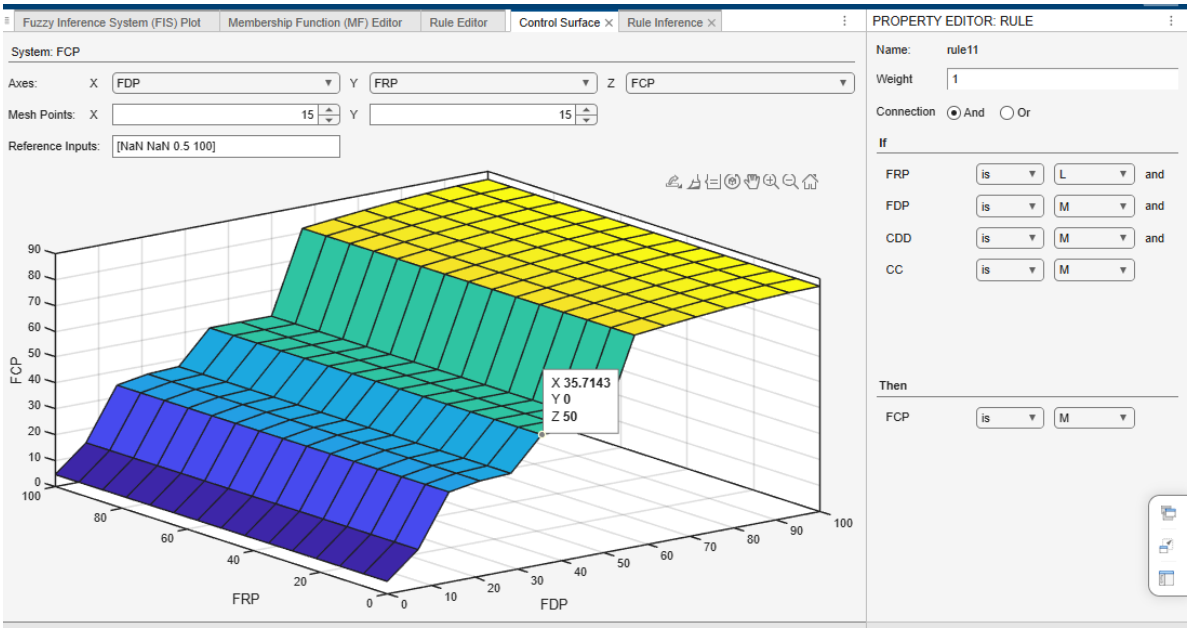


Fig: Control Surface Diagram of phase 3

X axis FDP is moderate (around 35) and Y axis FRP is low (0), Z axis the value of FCP is moderate (around 50), indicating a medium number of fault count prediction

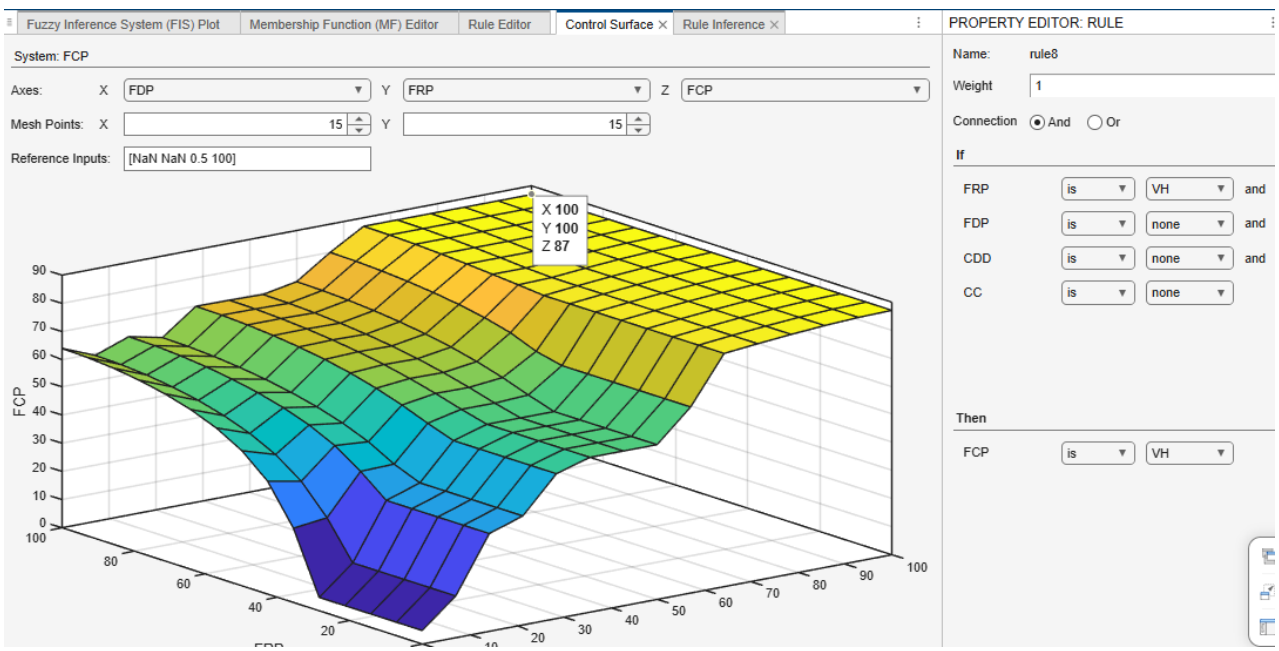


Fig: Control Surface Diagram of phase 3

X axis FDP is very high (100) and Y axis FRP is very high (100), Z axis the value of FCP is very high (around 87), indicating a very high number of fault count prediction

Observation from control surface:

1. **Increase in FDP** gives **Strong increase in FCP, Decrease in FDP** gives **Sharp decrease in FCP.**
2. **Increase in FRP** gives **Increase in FCP, Decrease in FRP** gives **Decrease in FCP**
3. **Both High** (FDP, FRP) → **FCP Very High** (maximum region)
4. **Both Low** (FDP , FRP) → **FCP Very Low** (minimum region)
5. The fuzzy rule base for FCP demonstrates that FDP is the most dominant factor influencing fault count, while FRP, CDD, and CC act as supporting parameters that amplify or moderate the final prediction through nonlinear rule interaction

5	If FDP is VH then FCP is VH	1	rule5
6	If FRP is M and FDP is M and CDD is M and CC is H then FCP is H	1	rule6
7	If FRP is M and FDP is M and CDD is H and CC is M then FCP is H	1	rule7
8	If FRP is VH and FDP is H and CDD is M and CC is M then FCP is VH	1	rule8
9	If FRP is L and FDP is L and CDD is L and CC is L then FCP is L	1	rule9
10	If FRP is M and FDP is L and CDD is M and CC is M then FCP is M	1	rule10
11	If FRP is L and FDP is M and CDD is M and CC is M then FCP is M	1	rule11
12	If FRP is H and FDP is M and CDD is M and CC is M then FCP is H	1	rule12
13	If FRP is L and FDP is H and CDD is M and CC is M then FCP is M	1	rule13
14	If FRP is VH and FDP is H and CDD is H and CC is H then FCP is VH	1	rule14
15	If FRP is M and FDP is VH and CDD is H and CC is H then FCP is VH	1	rule15
16	If FRP is H and FDP is M and CDD is M and CC is M then FCP is H	1	rule16
17	If FRP is H and FDP is M and CDD is M and CC is M then FCP is H	1	rule17
18	If FRP is H and FDP is L and CDD is M and CC is M then FCP is M	1	rule18
19	If FRP is H and FDP is L and CDD is M and CC is M then FCP is M	1	rule19
20	If FRP is VH and FDP is VH and CDD is H and CC is H then FCP is VH	1	rule20
21	If FRP is M and FDP is M and CDD is M and CC is M then FCP is M	1	rule21
22	If FRP is M and FDP is H and CDD is M and CC is M then FCP is H	1	rule22
23	If FRP is H and FDP is H and CDD is H and CC is M then FCP is VH	1	rule23

Fig: Rule Base for Requirement Phase – Fuzzy Rules Mapping (FRP, FDP, CDD, CC) to FCP

Error Analysis Results

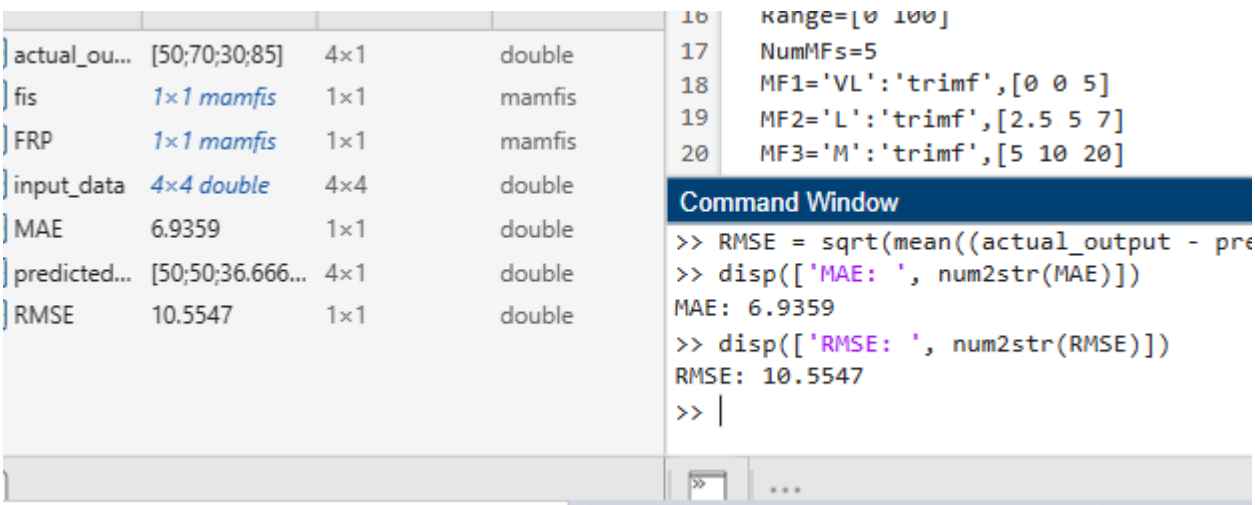
The performance of the fuzzy fault prediction model is evaluated using MATLAB. The Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are calculated for the final output variable (FCP).

The obtained results are:

- MAE = 6.94
- RMSE = 10.55

The low MAE value indicates that the average deviation between predicted and actual fault values is small. The RMSE value is slightly higher due to the penalization of larger errors, but still remains within an acceptable range.

These results demonstrate that the proposed fuzzy inference system provides accurate and reliable fault prediction.



actual_ou...	[50;70;30;85]	4×1	double
fis	1×1 mamfis	1×1	mamfis
FRP	1×1 mamfis	1×1	mamfis
input_data	4×4 double	4×4	double
MAE	6.9359	1×1	double
predicted...	[50;50;36.666...	4×1	double
RMSE	10.5547	1×1	double

```
Range=[0 100]
NumMFs=5
MF1='VL':'trimf',[0 0 5]
MF2='L':'trimf',[2.5 5 7]
MF3='M':'trimf',[5 10 20]
```

Command Window

```
>> RMSE = sqrt(mean((actual_output - pre
>> disp(['MAE: ', num2str(MAE)])
MAE: 6.9359
>> disp(['RMSE: ', num2str(RMSE)])
RMSE: 10.5547
>>
```

Fig: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) from FCP

Sensitivity Results

Sensitivity Analysis (FCP Model)

Sensitivity analysis is performed by introducing a small variation in each input parameter (FRP, FDP, CDD, and CC) and observing the change in output using MATLAB.

The results show that:

- FDP has a noticeable sensitivity value, indicating it significantly influences the final fault count
- FRP, CDD, and CC show low or negligible sensitivity at the selected input point, indicating their influence is limited in this region

This confirms that **Fault Density Prediction (FDP)** is the *most dominant factor* affecting **Fault count prediction**, while other parameters act as supporting factors depending on operating conditions.

```
>>base_input = [80 80 0.7 300];
```

```
>>Output =
```

```
Command Window
Sensitivity Results:
FRP: 0
FDP: 0.13257
CDD: 0
CC : 0
>>
```

Conclusion

The developed **Fuzzy Inference System (FIS)** effectively models software fault prediction across different phases by integrating key metrics such as **RCR, RIW, PM, FDP, CDD, and CC**. The system demonstrates that:

- **Requirement Change Rate (RCR)** is the most influential factor in determining **Fault Risk Prediction (FRP)**, with higher changes leading to significantly increased risk levels.
- **Fault Density Prediction (FDP)** plays a dominant role in determining **Fault Count Prediction (FCP)**, showing a strong direct relationship with the number of faults.
- **Code Defect Density (CDD)** and **Cyclomatic Complexity (CC)** act as supporting factors that **amplify fault occurrence**, especially when combined with high FDP and FRP.

The control surface analysis confirms that:

- The system exhibits **nonlinear behavior with smooth transitions**, ensuring realistic and stable predictions.
- **High input conditions consistently produce high fault risk and fault count**, while low inputs result in minimal faults, validating logical consistency.
- The rule base effectively captures **interdependencies among variables**, enabling accurate mapping from inputs to outputs.